Identification and validation of an explainable early-stage chronic kidney disease prediction model: a multicenter retrospective study

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Summary

Background Chronic Kidney Disease (CKD) has become a significant global public health issue, affecting approximately 10% of adults. Due to the lack of obvious symptoms in the early stages, CKD is often difficult to diagnose in a timely manner, leading to the gradual progression of the disease, which can eventually develop into End-Stage Renal Disease (ESRD). This study applied machine learning (ML) methods to integrate patient clinical data and developed an early CKD prediction model applicable to individuals without diabetes, hypertension, or coronary heart disease. The model aims to enhance the accuracy of early CKD risk assessment, thereby delaying disease progression and improving patient outcomes.

Methods This study is a retrospective multicenter study conducted in China, including patients with CKD and healthy individuals who underwent physical examinations from February 2021 to April 2024. Six ML methods, including Decision Tree, Multilayer Perceptron, and XGBoost, were used to predict CKD, integrating different combinations of features such as blood routine, urine analysis, and blood biochemistry. Multiple evaluation metrics, including AUC and F1 score, were used to compare the prediction performance. The SHAP interpretability method was applied to assess feature importance and explain the final model's results.

Findings Data from three hospitals were used in this study, with the dataset divided into training and internal validation sets (CKD: 11,436 cases, non-CKD: 10,004 cases) and an external validation set (CKD: 350 cases, non-CKD: 473 cases). Among the six ML models, XGBoost performed the best. Regarding feature combinations, the "blood routine + urinalysis + basic information" combination yielded the best performance (AUC = 0.9235, external validation AUC = 0.8962). Additionally, a web tool was developed in this study to facilitate the application of early CKD risk prediction in clinical practice.

Interpretation This study applied an interpretable ML model to effectively predict early CKD. Even when using the relatively low-cost "blood routine + urinalysis + basic information" combination, the model still demonstrated high prediction accuracy. This method has potential clinical application prospects and may help identify early CKD, reducing the risk of disease progression.

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Keywords: Chronic kidney disease; Early prediction; Machine learning; SHAP; Prediction model

Research in context

Evidence before this study

We conducted a search on PubMed for relevant literature published up to August 31, 2024, with no language restrictions, using "Chronic Kidney Disease" and "Prediction Model" as search terms, covering titles and abstracts. Although several studies on Chronic Kidney Disease (CKD) prediction have been published, none specifically focused on early CKD prediction models that combine physical examination and outpatient data, or used different feature combinations for tests.

Added value of this study

This study developed an interpretable CKD prediction model based on the XGBoost algorithm, which performed excellently in both internal and external validations. The model has been translated into a convenient application aimed at enhancing its practicality in clinical settings. Considering the cost of the tests, the "blood routine + urinalysis + basic information" feature combination was used for early CKD prediction, which still maintained high accuracy.

Implications of all the available evidence

The model developed in this study has the potential to become a practical diagnostic tool to help clinicians identify early patients with CKD. To facilitate clinical application, we have also developed a freely accessible web-based tool available at: https://github.com/Jiayi-Njucm/early-stage-ckd-prediction.

Introduction

Chronic Kidney Disease (CKD) is a progressive disease affecting over 10% of the global population and can eventually develop into End-Stage Renal Disease (ESRD).[1] Treatment primarily involves non-pharmacological strategies (such as diet and lifestyle modifications) and pharmacological interventions (such as RAAS inhibitors, SGLT2 inhibitors, etc.), while managing cardiovascular risks associated with CKD, preventing infections, and addressing acute kidney injury are also critical for improving patient prognosis.[2]

CKD is divided into five stages, with ESRD (Stage G5) being the most severe. In the early stages of CKD, despite the lack of noticeable symptoms, kidney function has already been impaired and is often overlooked. Once the disease progresses to later stages, patients typically require renal replacement therapy (such as dialysis or kidney transplantation), at which point medical costs rise significantly, often accompanied by the risk of cardiovascular diseases.[3,4] Therefore, early detection and timely intervention are crucial for delaying disease progression, reducing complications, and improving prognosis.[5] According to the 2024 KDIGO guidelines,[6] it is recommended to assess estimated glomerular filtration rate (eGFRcr) based on creatinine in adults at risk of CKD; when possible, a combined assessment of creatinine and cystatin C (eGFRcr-cys) is suggested for more accurate GFR classification.

With advancements in data science, machine learning (ML) methods such as Random Forest,[7] XGBoost,[8] and LightGBM[9] have shown great potential in early disease prediction. For example, ML has been used to predict pediatric acute kidney injury (AKI),[10] IgA nephropathy,[11] diabetes,[12] hyperuricemia,[13] among others. However, in CKD prediction research, most studies focus on identifying patients in the middle to late stages of the disease,[14–17] with relatively fewer studies targeting early CKD detection. One study combined the Eurygasters optimization algorithm with an ensemble deep learning model to propose an early detection method for CKD,[18] which, through feature selection, data normalization, and hyperparameter tuning, demonstrated superior detection accuracy on benchmark datasets compared to traditional models. However, sequence models such as Long Short-Term Memory (LSTM),[19] Bidirectional LSTM (BiLSTM),[20] and Bidirectional Gated Recurrent Units (BiGRU)[21] still face scalability and computational efficiency limitations when handling large-scale datasets.

Additionally, although ML methods possess strong predictive capabilities due to their complexity, they are often referred to as "black boxes"[22] because their internal mechanisms are difficult to interpret. Previous studies[16] have enhanced model interpretability by introducing multiple attribution algorithms. These algorithms not only help identify key features related to CKD progression (such as hematuria, proteinuria, and potassium levels), but also provide more intuitive data support for clinicians. However, some attribution methods (such as Gradient SHAP) have certain inconsistencies with existing clinical knowledge, indicating the need to further improve these models' interpretability and validate their effectiveness in real-world clinical applications. Feature-based interpretability methods[23,24] help address the "black-box" problem. This study uses SHAP (SHapley Additive exPlanations) for interpretability and visualizes individual variable predictions in ML models. SHAP, based on Shapley values from cooperative game theory, is used to quantify each participant's contribution to the overall payoff in a game. This concept is applied in the ML field to quantify the contribution of each feature to the model's prediction.

This study, based on a multicenter retrospective cohort dataset, aims to develop and validate an interpretable ML model for early-stage CKD, quantify key risk factors, and predict early-stage CKD (CKD stages 1–2). The model is designed as a web-based online tool to assist health checkup centers and other non-nephrology specialty outpatient clinics in identifying potential CKD risk patients for follow-up evaluation.

Methods

Study population

This multicenter retrospective study included data from three hospitals between February 2021 and April 2024: Jiangsu Province Hospital of Chinese Medicine, Suzhou Hospital of Integrated Traditional Chinese and Western Medicine, and Zhongda Hospital, Southeast University School of Medicine.

Data from the outpatient department of Jiangsu Province Hospital of Chinese Medicine and the physical examination center of Suzhou Hospital of Integrated Traditional Chinese and Western Medicine were used for model training and internal validation.

Data from the physical examination center of Jiangsu Province Hospital of Chinese Medicine and the nephrology outpatient department of Zhongda Hospital, Southeast University School of Medicine were used for external validation.

Ethical statement

This study was approved by the Institutional Review Boards of the participating hospitals (Approval number: 2022NL-198-02, Supplementary Appendix S1). Due to the retrospective study design and the use of deidentified patient information, the requirement for informed patient consent was waived.

Inclusion and exclusion criteria

Inclusion Criteria: Adults aged 18 and above; Complete laboratory examination data, including blood routine, urinalysis, and blood biochemistry; eGFR $\geq$ 60.

Exclusion Criteria: Diagnosis includes major diseases such as diabetes, hypertension, acute kidney injury, cancer, coronary heart disease, etc.; cases with complete missing data on age, gender, or creatinine.

Data collection and preprocessing

This study constructed models using different combinations of blood routine, urinalysis, and blood biochemistry, including only features that were tested across all research centers to ensure data consistency and comparability. Some features (such as granular casts, calcium, sodium, etc.) were only tested in certain hospitals, resulting in a high missing rate (over 25%) across the entire dataset. To reduce potential bias caused by cross-center data differences, these features were excluded from the analysis. Additionally, interactions between features were observed, and adjustments were made accordingly. For example, since PCT = MPV * PLT, we retained MPV and PLT; based on neutrophil count and lymphocyte count, the Neutrophil-Lymphocyte Ratio (NLR) was constructed. Given the significant impact of creatinine on eGFR, creatinine was excluded. Moreover, since ACR testing was not performed in the health checkup population, this feature was not included in the analysis.

A total of 26 variables were ultimately included: basic information consisted of gender and age; blood routine included white blood cell count, red blood cell count, platelets, hemoglobin, basophil percentage, eosinophil percentage, mean platelet volume (MPV), and NLR; urinalysis included vitamin C, ketone bodies, urinary protein, white blood cells, occult blood, glucose, bilirubin, and nitrites; blood biochemistry included total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, alanine transaminase (ALT), aspartate transaminase (AST), uric acid, triglycerides, and glucose.

For handling missing values, multiple imputation (MICE) was applied for continuous variables, while categorical variables were filled using the mode to mitigate the impact of missing data on the model.

CKD definition

According to the KDIGO 2024 guidelines[6] on CKD staging, CKD stages 1–2 are defined as: 60 $\leq$ eGFR <90 or eGFR $\geq$90 with urinary protein positivity. Non-CKD: eGFR $\geq$90 and negative urinary protein. CKD diagnosis is primarily staged (CKD1-5) based on eGFRcr-cys. GFR assessment relies on indicators such as serum creatinine, cystatin C, serum albumin, body weight, age, sex, and race, while the degree of proteinuria is not included in the staging evaluation. eGFR is calculated using the CKD-EPI creatinine formula.

Model development and comparison

During model development, the study data was divided into 60% training set, 20% validation set, and 20% test set. A predictive model was constructed using the aforementioned variables, employing six ML models: Decision Tree, Random Forest, Multilayer Perceptron (MLP), Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM) to predict early-stage CKD. Several commonly used evaluation metrics[25] were applied, including the area under the Receiver Operating Characteristic (ROC) curve (AUC), sensitivity, specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), accuracy, and F1 score, to assess the reliability of these models. The DeLong method was used to test the differences in AUC between models. Additionally, Youden's J index was calculated to comprehensively evaluate the model's performance in balancing sensitivity and specificity.

Model interpretability

SHAP is a technique used to explain ML model predictions. It is based on the Shapley value from game theory, applying this concept to ML to quantify each feature's contribution to the model prediction. The size of the SHAP value indicates the importance of a feature in the model prediction. Larger SHAP values mean that the feature has a greater impact on the prediction, and thus has higher importance in the model; conversely, smaller SHAP values indicate a lesser impact. Additionally, SHAP values can be either positive or negative. Positive values indicate that the feature has a positive impact on the prediction, increasing the predicted value, while negative values indicate a negative impact, lowering the predicted value.

Sensitivity and subgroup analyses

To evaluate the robustness and generalizability of the model, we conducted sensitivity analyses to assess model stability and subgroup analyses to examine its generalizability. The sensitivity analyses consisted of two components: First, we compared the model performance with and without the use of SMOTE to address class imbalance in the training set. Second, we assessed the impact of two missing data handling strategies, MICE and complete case analysis, on model performance.

Subgroup analyses were performed in the provincial cohort of the external validation set (n = 704, including both CKD and non-CKD patients). Stratification variables included commonly used clinical indicators such as sex, age, white blood cell count, red blood cell count, platelet count, and hemoglobin level, to evaluate the model's performance across different subpopulations.

Web application

To facilitate practical use, a Python Flask-based web application was developed to predict CKD risk based on the user's basic information, blood routine, and urinalysis. Furthermore, by integrating the SHAP interpretability analysis, the application provides feature importance information, helping users more intuitively understand the key factors affecting their risk assessment.

Statistical analysis

Data analysis was performed using R 4.4 (https://www.r-project.org/) and Python 3.10 (https://www.python.org). Continuous variables were presented as the median and interquartile range (IQR), and their distribution was assessed using the Kolmogorov–Smirnov (K–S) test. Comparisons of continuous variables were conducted using the Mann–Whitney U test or the Kruskal–Wallis H test, as appropriate. Categorical variables were expressed as counts with percentages and compared using the chi-square test or Fisher's exact test. A two-tailed P-value <0.05 was considered statistically significant.

Role of funding source

The funders of this study had no involvement in the study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Patient characteristics

The overall study design process is illustrated in Fig. 1.

In this study, we performed descriptive analysis on demographic and clinical characteristics for 11,436 patients with CKD and 10,004 healthy controls. Tables 1 and 2 provide detailed comparisons between the two groups in terms of demographic characteristics, blood routine, biochemical, and urine routine parameters.

Table 1 systematically describes and statistically tests the basic characteristics, blood routine, and biochemical indicators of patients with CKD (n = 11,436) and healthy controls (n = 10,004). The results show that the median age of the CKD group was 57 years (IQR: 43–69), while the median age of the non-CKD group was 56 years (IQR: 50–61), with a statistically significant difference in age distribution (P < 0.0001). Additionally, the proportion of males in the CKD group was 39.68%, higher than the 27.65% in the non-CKD group.

In terms of blood routine indicators, the CKD group had significantly different white blood cell count, red blood cell count, platelet count, and hemoglobin compared to the non-CKD group (P < 0.0001). Specifically, the platelet count was significantly lower in the CKD group, possibly due to reduced platelet production or increased destruction caused by declining kidney function. Furthermore, differences in basophil and eosinophil percentages suggest that inflammation or immune status changes are involved in the pathological process of CKD.

Regarding biochemical indicators, the CKD group had significantly higher uric acid levels compared to the non-CKD group (P < 0.0001), reflecting impaired uric acid excretion due to kidney dysfunction, with many patients with CKD also having hyperuricemia. Meanwhile, LDL-C levels were lower, and triglyceride levels were higher in the CKD group (P < 0.0001), indicating lipid metabolism abnormalities. Additionally, ALT and AST levels were slightly higher in the CKD group than in the non-CKD group (P < 0.0001), which may indicate liver function involvement.

In terms of urine tests (Table 2), only 26.77% of patients with CKD had negative urine protein, while all healthy individuals had negative urine protein (P < 0.0001), further confirming that urine protein is a core biomarker for diagnosing CKD.

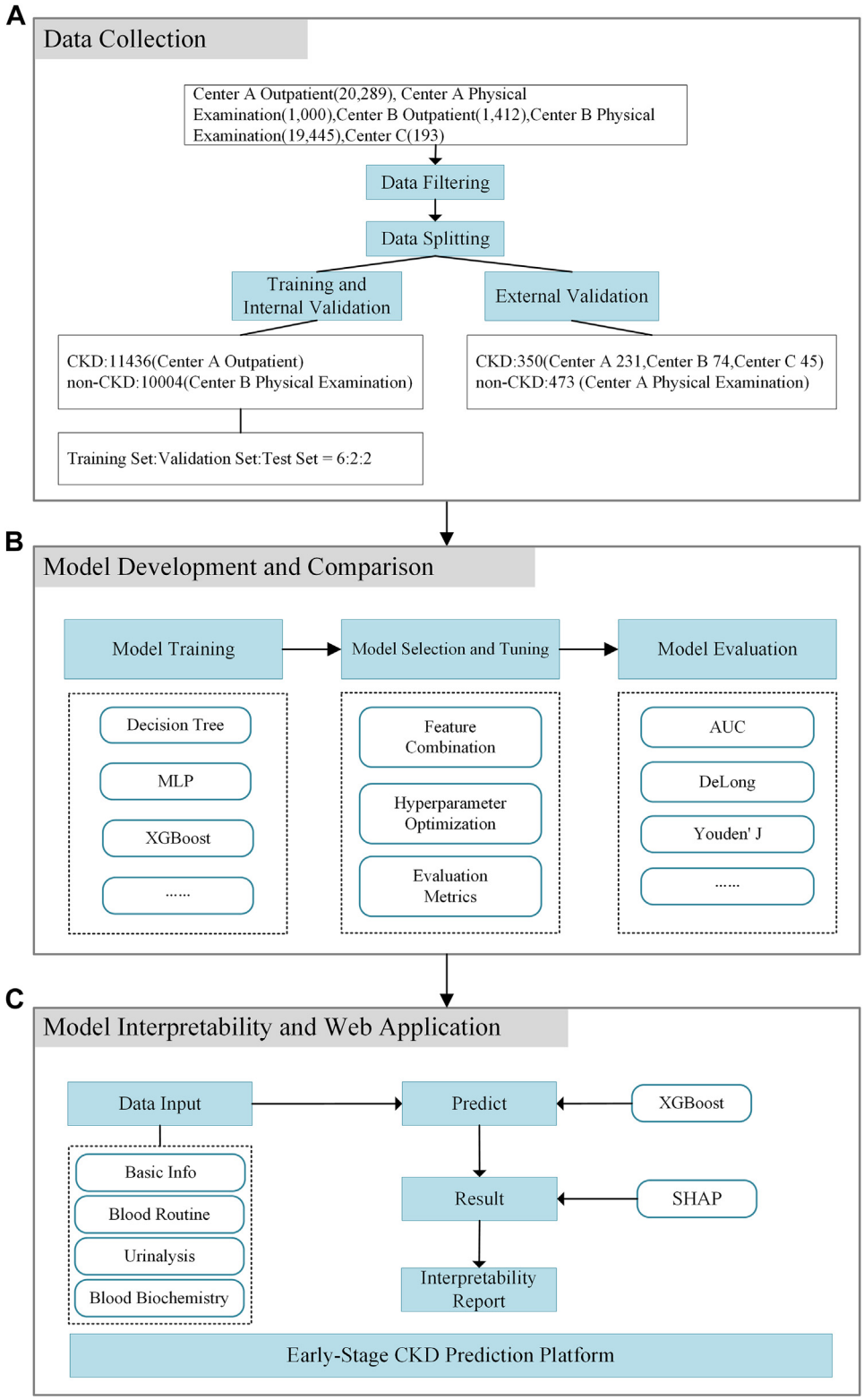


Fig. 1: **Research Flowchart**. **(A)** Data from patients with CKD and healthy individuals were obtained from three hospitals for model training, internal validation, and external validation. **(B)** Model Development and Comparison: Various machine learning models were trained, and the

Type	Feature	CKD (n = 11,436)	Non-CKD (n = 10,004)	P
Basic	Age, year	58 (44, 68)	57 (51, 62)	<0.0001
	Male, n (%)	4538/11,436 (39.68)	2766/10,004 (27.65)	
Blood Routine	WBC, 10^9/L	6.06 (5.09, 7.3)	5.91 (5.02, 6.95)	<0.0001
	RBC, 10^12/L	4.58 (4.24, 4.97)	4.53 (4.3, 4.83)	<0.0001
	PLT, 10^9/L	217 (180, 257)	226 (193, 265)	<0.0001
	HGB, g/L	138 (128, 149)	138 (131, 148)	<0.0001
	BASO%, n (%)	0.4 (0.3, 0.6)	0.4 (0.3, 0.6)	<0.0001
	EOS%, n (%)	1.8 (1.1, 2.9)	1.8 (1.2, 2.8)	0.0052
	MPV, fL	10 (9.4, 10.7)	10.2 (9.5, 11.1)	<0.0001
	NLR	1.61 (1.05, 2.25)	1.56 (1.23, 1.98)	0.9662
Blood Biochemistry	GLU, mmol/L	5.33 (4.97, 5.83)	5.25 (4.88, 5.82)	<0.0001
	TC, mmol/L	5 (4.28, 5.74)	4.91 (4.31, 5.54)	<0.0001
	HDL-C, mmol/L	1.39 (1.19, 1.64)	1.4 (1.18, 1.65)	0.8146
	LDL-C, mmol/L	2.81 (2.26, 3.4)	3 (2.44, 3.61)	<0.0001
	ALT, U/L	19 (14, 28)	17 (13, 25)	<0.0001
	AST, U/L	20 (17, 25)	20 (17, 24)	<0.0001
	UA, μmol/L	333 (280, 395)	284.6 (241, 336.83)	<0.0001
	TG, mmol/L	1.39 (1, 2.01)	1.33 (0.94, 1.92)	<0.0001

Continuous values were presented as median [interquartile range]. Categorical values were presented as number (percentage). WBC: White Blood Cell Count; RBC: Red Blood Cell Count; PLT: Platelet Count; HGB: Hemoglobin; BASO%: Basophil Percentage; EOS%: Eosinophil Percentage; MPV: Mean Platelet Volume; NLR: Neutrophil to Lymphocyte Ratio; GLU: Glucose; TC: Total Cholesterol; HDL-C: High-Density Lipoprotein Cholesterol; LDL-C: Low-Density Lipoprotein Cholesterol; ALT: Alanine Aminotransferase; AST: Aspartate Aminotransferase; UA: Uric Acid; TG: Triglycerides.

Table 1: **Descriptive statistics of basic information, and continuous feature.**

Model Development and Comparison

In this study, we adopted the following model training and evaluation process:

① Data Preprocessing: First, the data is loaded and missing values are handled using methods such as multiple imputation. The dataset is split into training, validation, and test sets using stratified sampling in a 6:2:2 ratio, respectively, ensuring consistent distribution of classes across training and test datasets.

② Model Training: The model is initialized and trained on the training set. Cross-validation is used to evaluate model performance, helping to assess the model's stability.

③ Model Selection and Tuning: Based on the results of cross-validation, the best-performing model is selected. Hyperparameters are then tuned using methods such as grid search or random search. The optimized model is evaluated on the validation set, and performance metrics for the validation set are output.

④ Model Evaluation: The model's performance is further evaluated using the validation set, including metrics such as accuracy, precision, recall, F1 score, and Youden's J statistic. The classification threshold is adjusted to optimize the balance between precision and recall. A statistical significance test (DeLong test) is also performed to ensure the statistical significance of performance differences.

⑤ SHAP Interpretability Analysis: SHAP is used to analyze the contribution of each feature to the model's predictions, helping to explain the model's decision-making process. SHAP values are visualized to gain a deeper understanding of feature importance.

⑥ External Validation: The model's generalizability is validated using an external dataset, and evaluation metrics such as accuracy are calculated to ensure the model's practical performance in real-world applications.

This study selected six ML models for CKD prediction: Decision Tree, Random Forest, MLP, AdaBoost, XGBoost, and LightGBM. The ROC curves, PRC curves, and confusion matrices of each model are shown in Figs. 2 and 3. The evaluation metrics include: Accuracy, F1 score, Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), AUC, and PR-AUC.

From the evaluation metrics in Table 3, XGBoost and LightGBM performed excellently in terms of AUC and F1 score. Specifically, XGBoost achieved an AUC of 0.9397 and an F1 score of 0.8690, while LightGBM had an AUC of 0.9400 and an F1 score of 0.8729, both showing strong classification ability. The Decision Tree performed relatively poorly in these two metrics, with an AUC of 0.8455 and an F1 score of 0.7359, indicating lower sensitivity and higher specificity. Random Forest showed improvement over Decision Tree, with an AUC of 0.9282 and an F1 score of 0.8551, exhibiting a more balanced sensitivity and specificity. The MLP had an AUC of 0.8979 and an F1 score of 0.7992, performing moderately with high sensitivity. AdaBoost achieved an AUC of 0.8740 and an F1 score of 0.8055, performing similarly to MLP but with lower sensitivity.

Table 4 uses the DeLong test to perform a statistical comparison of the AUCs of different models to assess

best model was selected based on evaluation metrics, feature combinations, and parameter optimization. The model's performance and generalization ability were assessed through internal and external validation. **(C)** Model Interpretability and Web Application: SHAP values were used to analyze the impact of features on CKD prediction, providing interpretability for model decisions. A Python Flask-based application was developed to allow users to input test results for early-stage CKD risk prediction and generate an interpretability analysis report. **Center A:** Jiangsu Province Hospital of Chinese Medicine, **Center B:** Suzhou Hospital of Integrated Traditional Chinese and Western Medicine, **Center C:** Zhongda Hospital, Southeast University School of Medicine.

the significance of differences in model performance. XGBoost shows a significant advantage over all other models ($P < 0.0001$), especially compared to the Decision Tree, with a highly significant statistical difference. The AUC difference between XGBoost and LightGBM is not significant, indicating that both models have similar predictive abilities.

Optimal model

Among the six models mentioned above, based on the results of the DeLong test, the AUC difference between XGBoost and LightGBM is statistically insignificant ($P = 0.9624$), indicating similar AUC performance between the two. However, the AUC difference between XGBoost and other models (such as Decision Tree, MLP, and AdaBoost) is significant ($P < 0.0001$), showing the superiority of XGBoost over most models. In addition, XGBoost performs better in preventing overfitting, especially with larger datasets, where its regularization mechanism effectively enhances the model's generalization ability. Moreover, XGBoost offers more support in feature importance analysis and model interpretability, providing higher transparency and room for optimization in practical applications. Therefore, considering all factors, XGBoost was selected as the optimal model in this study. The tuned model parameters are as follows: random_state = 42, learning_rate = 0.01, max_depth = 5, min_child_weight = 5, n_estimators = 150, subsample = 0.8, colsample_bytree = 0.8, gamma = 0.1, reg_lambda = 3, reg_alpha = 0.5, eval_metric = "logloss".

Comparisons were made for different feature combinations to evaluate their performance in CKD prediction. The features used include blood routine, urine routine, biochemical indicators, and basic information. The results are shown in Figs. 4 and 5 and Table 5. For single feature categories, biochemical indicators (XCG) outperformed blood routine (NCG) and urine routine (XSH) when used alone. When combining two feature categories, XCG + NCG + Basic (blood routine + urine routine + basic information) performed better than other combinations, indicating a strong complementary effect between blood routine and urine routine. The full feature combination (XCG + NCG + XSH + Basic) showed the best performance, significantly improving the model's accuracy, sensitivity, and AUC.

The XGBoost (XCG + NCG + Basic) feature combination demonstrates stable performance in early CKD prediction tasks, making it suitable for early screening and risk assessment in healthcare environments with limited medical resources. It also facilitates follow-up visits for patients with potential CKD risks in health checkup centers or non-nephrology outpatient clinics. Table 6 shows the predictive performance of the XGBoost model on the test set using blood routine, urine routine, and basic information feature combinations. The AUC reached 0.9143, and the PR-AUC was 0.9415, indicating strong discriminative ability in distinguishing patients with CKD from healthy individuals. Overall classification performance was excellent, with accuracy (0.8561) and F1 score (0.8508) both being high, demonstrating a good balance between positive and negative samples. Sensitivity (0.7691) shows strong recognition capability for patients with CKD, while specificity (0.9555) indicates a high recognition rate for healthy individuals, with a low false positive rate. Furthermore, the positive predictive value (PPV = 0.9518) and negative predictive value (NPV = 0.7836) further confirm the reliability of the model's predictions. Overall, this model performs consistently in CKD prediction tasks and can be used for early screening and risk assessment.

Feature	0: Negative 1: +/± 2: ++, +++	CKD (n = 11,436)	Non-CKD (n = 10,004)	P
Vit C	0	6043/11,436 (52.84)	9675/10,004 (96.71)	<0.0001
	1	250/11,436 (2.19)	257/10,004 (2.57)	
	2	306/11,436 (2.68)	72/10,004 (0.72)	
KET	0	6713/11,436 (58.7)	9829/10,004 (98.25)	<0.0001
	1	170/11,436 (1.49)	168/10,004 (1.68)	
	2	48/11,436 (0.42)	7/10,004 (0.07)	
PRO	0	3061/11,436 (26.77)	10,004/10,004 (100)	<0.0001
	1	2536/11,436 (22.18)	0/10,004 (0)	
	2	1334/11,436 (11.66)	0/10,004 (0)	
LEU	0	4510/11,436 (39.44)	8266/10,004 (82.63)	<0.0001
	1	1608/11,436 (14.06)	891/10,004 (8.91)	
	2	830/11,436 (7.26)	847/10,004 (8.47)	
BLD	0	2787/11,436 (24.37)	7139/10,004 (71.36)	<0.0001
	1	2403/11,436 (21.01)	2111/10,004 (21.1)	
	2	1755/11,436 (15.35)	754/10,004 (7.54)	
UGLU	0	6428/11,436 (56.21)	9526/10,004 (95.22)	<0.0001
	1	158/11,436 (1.38)	134/10,004 (1.34)	
	2	345/11,436 (3.02)	344/10,004 (3.44)	
BIL	0	6806/11,436 (59.51)	9972/10,004 (99.68)	<0.0001
	1	107/11,436 (0.94)	24/10,004 (0.24)	
	2	18/11,436 (0.16)	8/10,004 (0.08)	
NIT	0	6707/11,436 (58.65)	9800/10,004 (97.96)	<0.0001
	1	100/11,436 (0.87)	204/10,004 (2.04)	
	2	124/11,436 (1.08)	0/10,004 (0)	

Continuous values were presented as median [interquartile range]. Categorical values were presented as number (percentage). Vit C: Vitamin C; KET: Ketones; PRO: Protein; LEU: Leukocytes; BLD: Occult Blood; UGLU: Urine Glucose; BIL: Bilirubin; NIT: Nitrite.

Table 2: **Descriptive statistics of urinalysis categorical feature.**

Model explainability

Fig. 6 presents the bar chart of SHAP values, used to quantify and visualize the contribution of each feature to the model's prediction. SHAP values indicate the degree to which each feature influences the final prediction result under different feature combinations. High

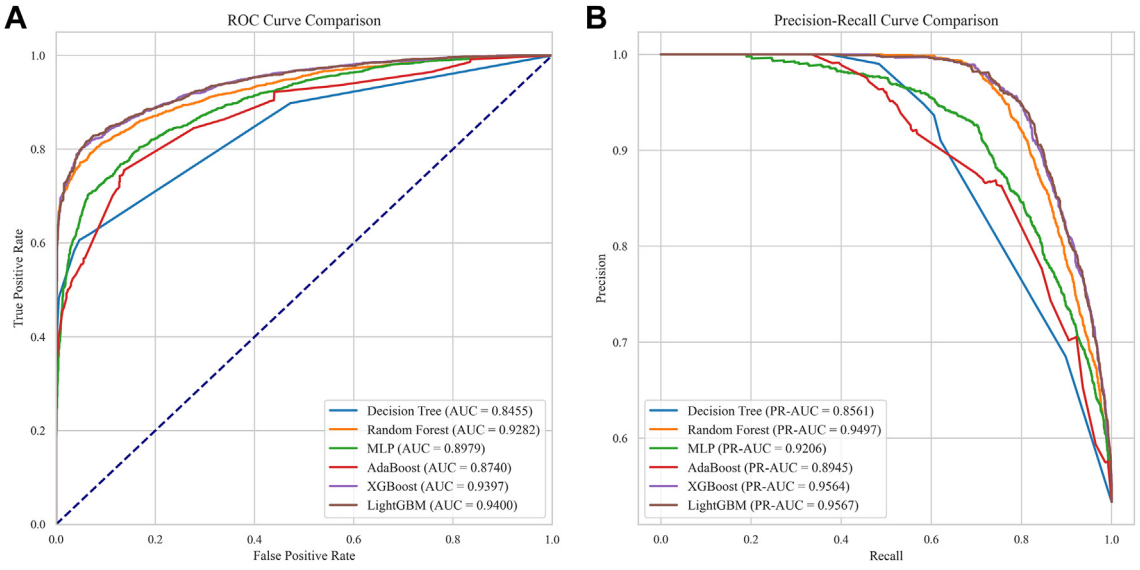


***Fig. 2:* Performance of ML models to predict CKD.** **(A)** ROC curves of the six ML models. **(B)** PRC Curves of the six ML models.

SHAP values indicate that the feature has a larger impact on the model's prediction.

Creatinine, as a core indicator for clinically assessing CKD stages, is an important biomarker in CKD progression.[26] In the absence of creatinine data, other features still provide predictive power. From Fig. 6A, it is evident that PRO (protein) and AGE (age) are the two most important features in the model's prediction, contributing far more to the final prediction than other features. In contrast, SEX, RBC count, and some other features have less influence. This suggests that proteinuria and age play a key role in predicting early-stage CKD in this model, while factors like sex have a weaker impact. Fig. 6C shows the SHAP values of all features.

Fig. 6B presents a cluster analysis that groups highly correlated features together, simplifying the feature

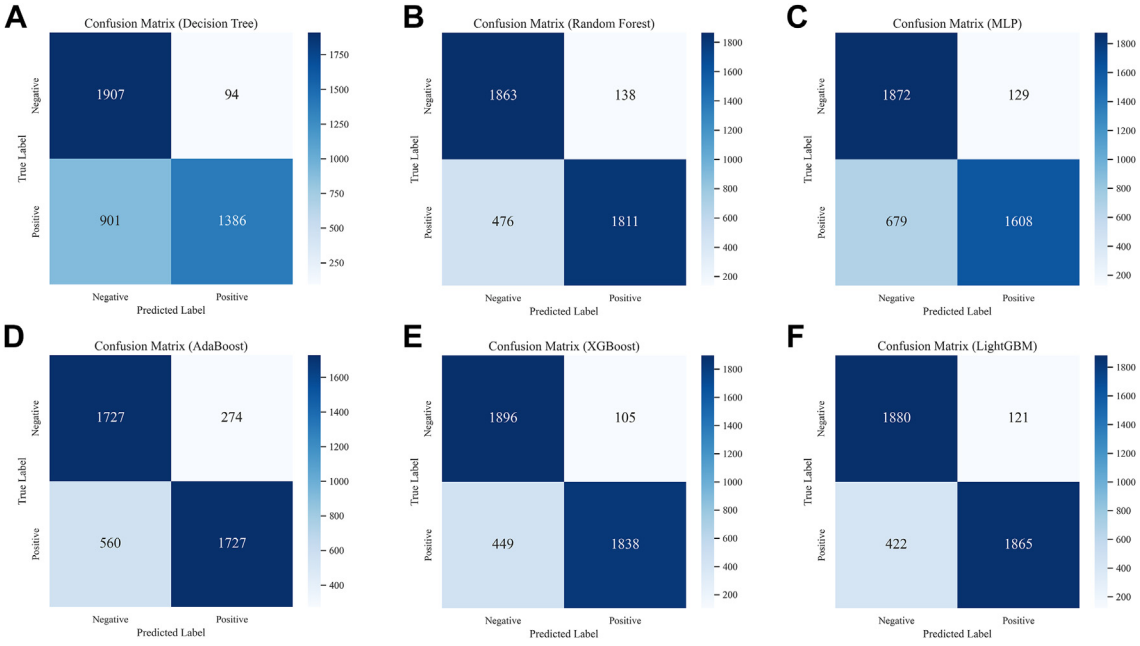


***Fig. 3:* Confusion Matrices of ML models.** **(A)** Decision Tree's Confusion Matrices. **(B)** Random Forest's Confusion Matrices. **(C)** MLP's Confusion Matrices. **(D)** AdaBoost's Confusion Matrices. **(E)** XGBoost's Confusion Matrices. **(F)** LightGBM's Confusion Matrices.

Model	Youden's J	Threshold	Accuracy	F1 score	Sensitivity	Specificity	PPV	NPV	AUC	PR-AUC
Decision Tree	0.5591	0.7585	0.7680	0.7359	0.6060	0.9530	0.9365	0.6791	0.8455	0.8561
Random Forest	0.7229	0.502	0.8568	0.8551	0.7919	0.9310	0.9292	0.7965	0.9282	0.9497
MLP	0.6386	0.6831	0.8116	0.7992	0.7031	0.9355	0.9257	0.7338	0.8979	0.9206
AdaBoost	0.6182	0.4736	0.8055	0.8055	0.7551	0.8631	0.8631	0.7551	0.8740	0.8945
XGBoost	0.7512	0.5249	0.8708	0.8690	0.8037	0.9475	0.9460	0.8085	0.9397	0.9564
LightGBM	0.755	0.5	0.8734	0.8729	0.8155	0.9395	0.9391	0.8167	0.9400	0.9567

MLP: Multi-Layer Perceptron; AUC: Area Under the ROC Curve; PPV: Positive Predictive Value; NPV: Negative Predictive Value; PR-AUC: Area Under the Precision-Recall Curve.

Table 3: **Model evaluation metrics.**

space. The combined feature SHAP value average for HGB + BASO% + MPV is 0.4, indicating that the combination of these blood indicators has strong predictive power in the model. This clustering result reveals the potential correlations between features, aiding in the understanding of biological mechanisms and supporting clinical decision-making.

Fig. 7A shows a beeswarm plot that displays the SHAP value distribution of each feature, providing a visual representation of how each feature influences the model's prediction results across different value ranges. SHAP values are color-coded (blue for lower feature values, purple for higher feature values) to reveal the relationship between feature values and prediction outcomes. Among all features, PRO (protein-related indicators) has the highest SHAP influence, with higher values (shown in pink) corresponding to higher SHAP values, indicating that higher PRO levels have a stronger impact on the model's predictions. AGE (age) is also an important feature, with a broad SHAP value distribution, and higher values (older age) tend to increase the model's prediction output. Additionally, blood-related features such as NLR, BASO%, PLT, MPV, and RBC_count also influence the prediction results, while features like SEX (gender) contribute less to the model's predictions. These results suggest that the model mainly relies on PRO, AGE, and some blood-related indicators, with factors like gender having less influence.

Model 1	Model 2	AUC1	AUC2	P-value
Decision Tree	Random Forest	0.8455	0.9282	P < 0.0001
Decision Tree	MLP	0.8455	0.8979	P < 0.0001
Decision Tree	AdaBoost	0.8455	0.874	0.0071
Decision Tree	XGBoost	0.8455	0.9397	P < 0.0001
Decision Tree	LightGBM	0.8455	0.94	P < 0.0001
Random Forest	MLP	0.9282	0.8979	0.0005
Random Forest	AdaBoost	0.9282	0.874	P < 0.0001
Random Forest	XGBoost	0.9282	0.9397	0.013
Random Forest	LightGBM	0.9282	0.94	0.1184
MLP	AdaBoost	0.8979	0.874	0.0145
MLP	XGBoost	0.8979	0.9397	P < 0.0001
MLP	LightGBM	0.8979	0.94	P < 0.0001
AdaBoost	XGBoost	0.874	0.9397	P < 0.0001
AdaBoost	LightGBM	0.874	0.94	P < 0.0001
XGBoost	LightGBM	0.9397	0.94	0.9624

MLP: Multi-Layer Perceptron; AUC: Area Under the ROC Curve.

Table 4: **The AUC DeLong test results.**

Fig. 7B presents scatter plots of SHAP values for each feature, revealing how each feature affects the model's prediction output across different value ranges. As shown, PRO and AGE, when their values increase, have a significant positive effect on the model's risk prediction, indicating that higher values of these features are typically associated with higher CKD risk. In contrast, BASO% and PLT show negative correlations, demonstrating the complex influence of these blood indicators on the model output under different physiological conditions.

Through SHAP value analysis of individual features, it becomes clear how the model adjusts its risk predictions based on changes in specific feature values. This detailed feature analysis not only enhances the model's interpretability but also provides profound insights into the role of specific biomarkers in CKD risk assessment, offering valuable references for clinical applications.

Fig. 8A and B are waterfall plots that show the contribution of different features to the model's prediction output (SHAP values) for a given patient. Each bar represents a feature, with blue indicating a negative contribution to the prediction result (lowering risk) and red indicating a positive contribution (increasing risk). Fig. 8C and D are force plots that illustrate how each feature contributes to the final prediction value f(x), starting from the model's baseline prediction value (base value). Blue arrows represent features that lower the prediction value, while red arrows represent features that increase the prediction value.

Fig. 8A and C correspond to the same individual (non-CKD). The features PRO (proteinuria), BASO%, and AGE (age) are the main factors contributing to a decrease in risk, significantly lowering the model's prediction value, suggesting a low CKD risk for the patient. Meanwhile, MPV and PLT are risk-increasing factors, though their contributions are relatively small.

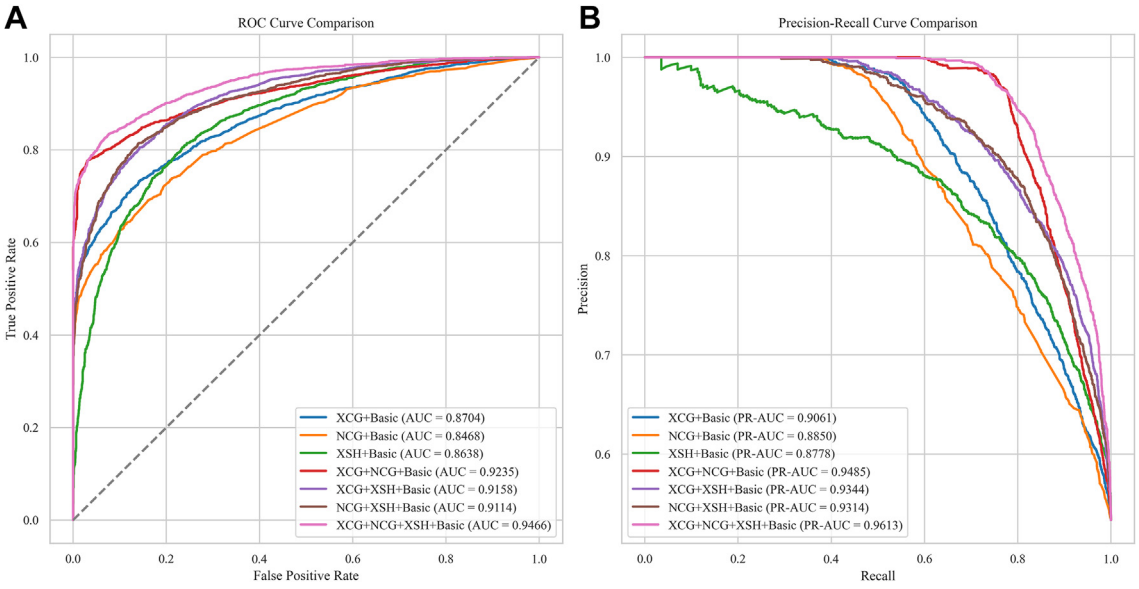


Fig. 4: **Performance of XGBoost Model with Different Feature Combinations**. **(A)** ROC curves of XGBoost. **(B)** PRC Curves of XGBoost. XCG: Blood Routine; XSH: Blood Biochemistry; NCG: Urinalysis.

The final prediction value is –0.547, indicating a low CKD risk for this patient.

Fig. 8B and D correspond to another individual (CKD). The PRO (proteinuria) level is significantly elevated, which causes a large increase in the model's prediction value. The final prediction value is 2.166, indicating an extremely high CKD risk for this patient.

External validation

The external validation dataset consists of independent cohorts from three different hospitals: the Provincial Central cohort (n = 704), the Zhongda cohort (n = 45), and the Suzhou cohort (n = 74), as shown in Tables 7 and 8. Among these, the Provincial Central cohort includes non-CKD (n = 473) and CKD (n = 231) cases, while the Zhongda and Suzhou cohorts are both CKD cases. This dataset is used for further analysis of the consistency of the model validation across different external datasets and their potential impact on CKD prediction.

Fig. 9 and Table 9 show that the model performs well in the external validation dataset, demonstrating excellent performance in identifying positive samples (high recall rate and sensitivity) and distinguishing between positive and negative samples (high ROC AUC and PR AUC).

Sensitivity and subgroup analyses

Supplementary Table S1 presents the comparison of model performance with and without the use of SMOTE under the final model (XGBoost) and feature combination (blood routine + urinalysis + basic information). The results show that the model's performance metrics under both strategies are highly comparable. The AUC on the test set was 0.9143 without SMOTE and 0.9134 with SMOTE, showing only a minimal difference. The accuracy was 0.8561 vs. 0.8554, and the F1 score was 0.8508 vs. 0.8498, both remaining at a high level. Other key metrics, such as Youden's J index, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV), also showed no substantial changes. These findings suggest that the model demonstrates strong robustness in handling class imbalance in the training set, and the use of SMOTE does not significantly affect the final model performance.

Supplementary Table S2 presents the comparison of model performance under two missing data handling strategies, MICE and complete case analysis, using the same test set. The results show that both methods achieved relatively similar AUC values, maintaining high model performance overall. Notably, the optimal classification threshold in the complete case analysis (0.1934) was lower than that under the MICE approach (0.5615). This may be related to the reduced sample size in the complete case dataset (n = 11,701, including 2227 positive and 9474 negative cases), which likely led to overall lower predicted probabilities for the positive class, thereby lowering the optimal threshold selected to maximize Youden's J index. Overall, the differences in model performance between the two strategies were minimal, indicating that the model is robust to the choice of missing data handling method.

Supplementary Table S3 presents the results of the subgroup analysis in the external validation set. The provincial cohort (n = 704) includes both patients with CKD (n = 231) and non-CKD individuals (n = 473),

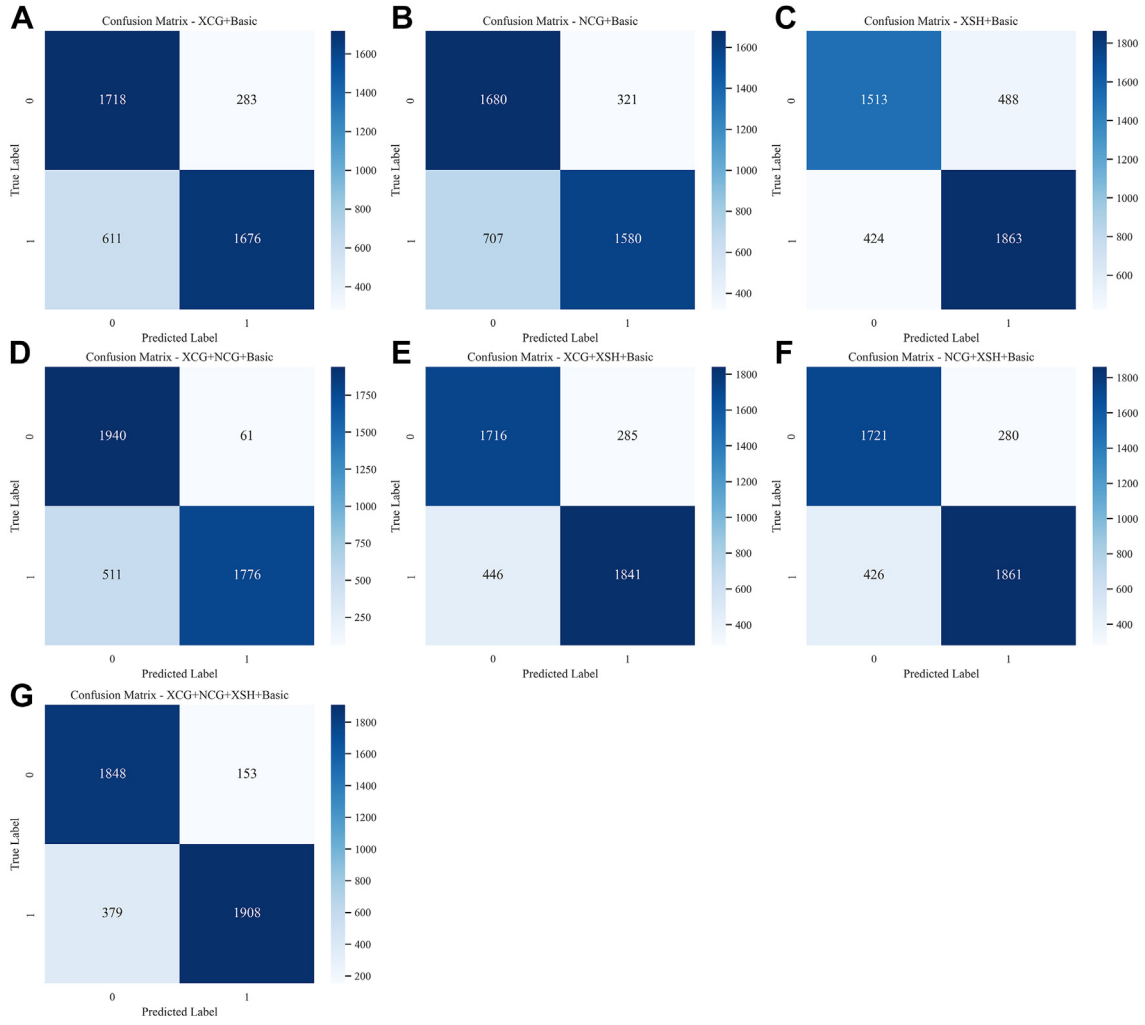


Fig. 5: **Confusion matrices of XGBoost model with different feature combinations**. **(A)** XGBoost confusion matrix for blood routine + basic information. **(B)** XGBoost confusion matrix for urinalysis + basic information. **(C)** XGBoost confusion matrix for blood biochemistry + basic information. **(D)** XGBoost confusion matrix for blood routine + urinalysis + basic information. **(E)** XGBoost confusion matrix for blood routine + blood biochemistry + basic information. **(F)** XGBoost confusion matrix for urinalysis + blood biochemistry + basic information. **(G)** XGBoost confusion matrix for blood routine + urinalysis + blood biochemistry + basic information.

Model	Youden' J	Threshold	Accuracy	F1 Score	Sensitivity	Specificity	PPV	NPV	AUC	PR-AUC
XCG + Basic	0.5914	0.5056	0.7915	0.7894	0.7328	0.8586	0.8555	0.7377	0.8704	0.9061
NCG + Basic	0.5304	0.4995	0.7603	0.7545	0.6909	0.8396	0.8311	0.7038	0.8468	0.885
XSH + Basic	0.5707	0.5042	0.7873	0.8034	0.8146	0.7561	0.7924	0.7811	0.8638	0.8778
XCG + NCG + Basic	0.7461	0.5898	0.8666	0.8613	0.7766	0.9695	0.9668	0.7915	0.9235	0.9485
XCG + XSH + Basic	0.6626	0.4963	0.8295	0.8344	0.805	0.8576	0.8659	0.7937	0.9158	0.9344
NCG + XSH + Basic	0.6738	0.4942	0.8354	0.8406	0.8137	0.8601	0.8692	0.8016	0.9114	0.9314
XCG + NCG + XSH + Basic	0.7578	0.4582	0.8759	0.8776	0.8343	0.9235	0.9258	0.8298	0.9466	0.9613

MLP: Multi-Layer Perceptron; AUC: Area Under the ROC Curve; PPV: Positive Predictive Value; NPV: Negative Predictive Value; PR-AUC: Area Under the Precision-Recall Curve.

Table 5: **Comparison of XGBoost model performance across different feature combinations.**

Model	Youden' J	Threshold	Accuracy	F1 Score	Sensitivity	Specificity	PPV	NPV	AUC	PR-AUC
XGBoost (XCG + NCG + Basic)	0.7247	0.5615	0.8561	0.8508	0.7691	0.9555	0.9518	0.7836	0.9143	0.9415

AUC: Area Under the ROC Curve; PPV: Positive Predictive Value; NPV: Negative Predictive Value; PR-AUC: Area Under the Precision-Recall Curve.

Table 6: **Performance of XGBoost (XCG + NCG + Basic) on the test set.**

making it suitable for calculating discrimination metrics such as AUC. In contrast, the Zhongda cohort (n = 45) and the Suzhou cohort (n = 74) consist solely of patients with CKD and, due to the lack of non-CKD controls, are not suitable for directly evaluating the model's discriminative performance (e.g., AUC). Given this data structure, calculating AUC separately for each external dataset could lead to biased or misleading results. Therefore, we selected the provincial cohort—containing both CKD and non-CKD individuals—for further subgroup sensitivity analyses to assess the model's robustness across different populations. In the vast majority of subgroups, the model demonstrated good predictive performance, with AUC values generally exceeding 0.90, indicating strong robustness and generalizability across diverse populations. It is worth noting that the AUC was relatively lower in the 45–60 years age subgroup (AUC = 0.7011), which can be attributed to the limited number of CKD cases in this group (n = 11). By contrast, in the more balanced 60–75 years subgroup, the model performed notably better, with an AUC of 0.8723.

Clinical application

A web application was developed using the Python Flask framework, allowing users to input personal information such as age, gender, as well as blood routine and urine routine data for CKD prediction, as shown in

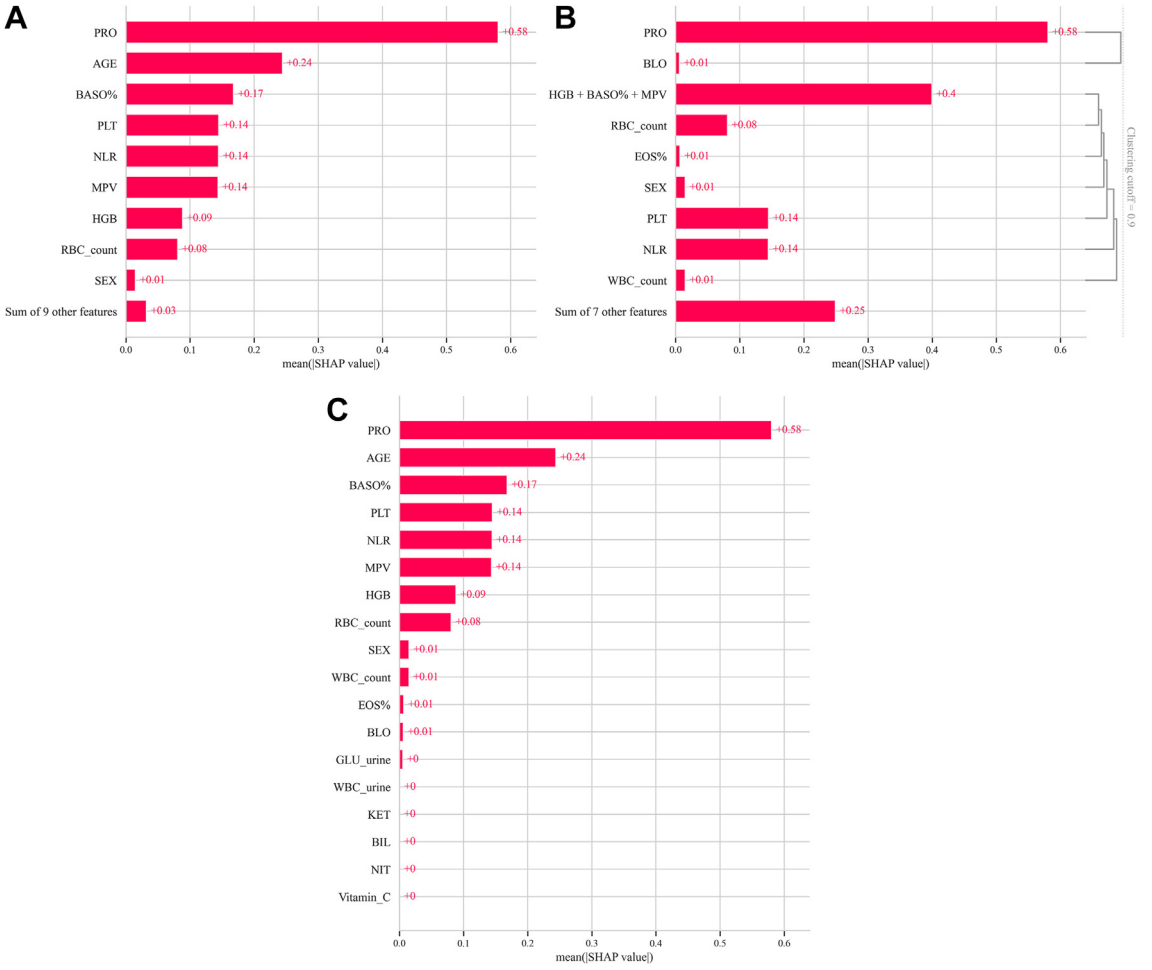


Fig. 6: **SHAP feature importance**. **(A)** Bar chart of the top 9 features. **(B)** Bar chart with clustering analysis. **(C)** Bar chart of the all features.

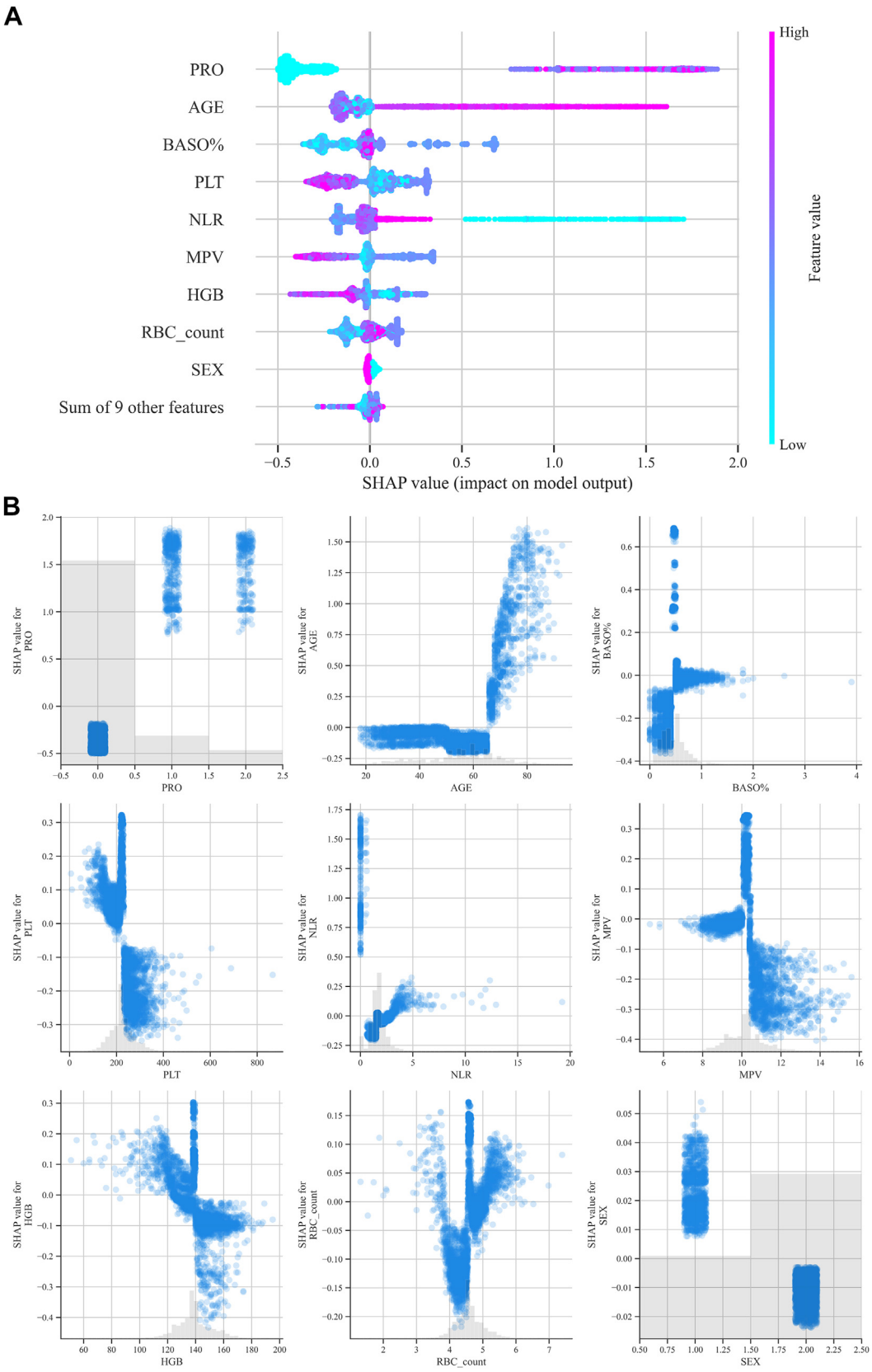


***Fig. 7*: Global model explanation by the SHAP method. (A)** Beeswarm plot. **(B)** Scatter plot.

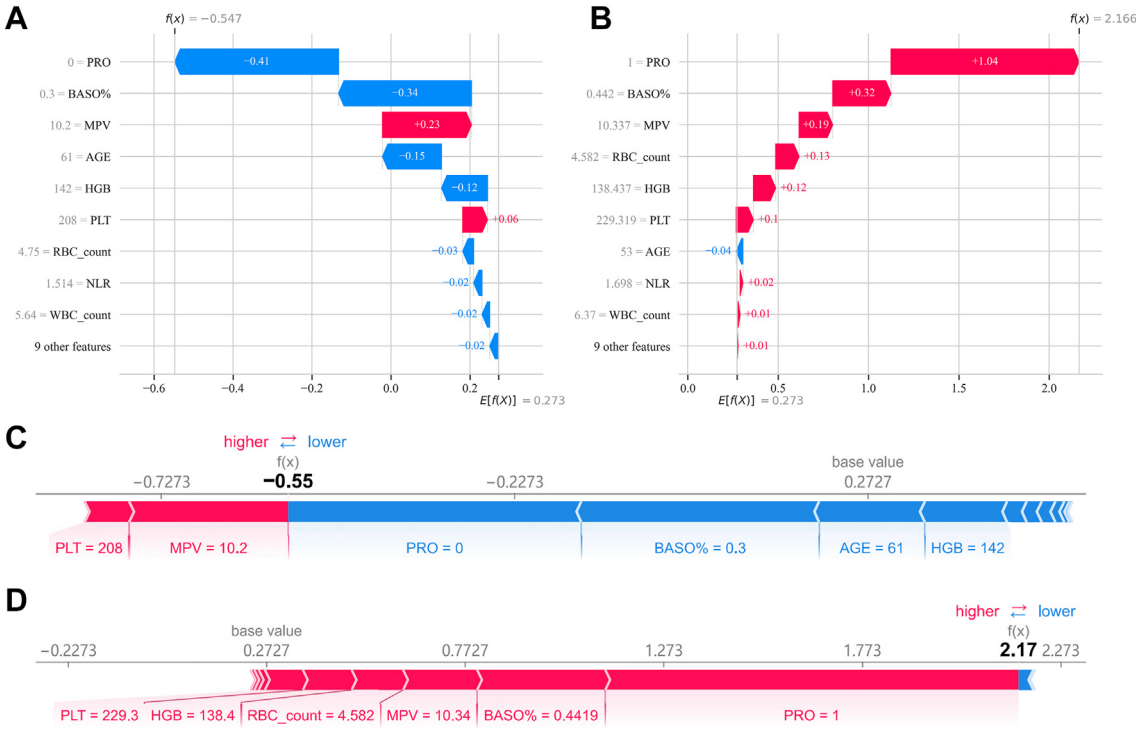


Fig. 8: **Local model explanation by the SHAP method.** **(A)** Waterfall plot for one non-CKD patient. **(B)** Waterfall plot for one CKD patient. **(C)** Force plot for one non-CKD patient. **(D)** Force plot for one CKD patient.

Feature	Center A (n = 704)	Center B (n = 74)	Center C (n = 45)
Age, year	56 (43, 69)	62.5 (48, 71.75)	59 (45, 68.5)
Male, n (%)	325/704 (46.16)	55/74 (74.32)	35/45 (77.78)
White Blood Cell Count, 10^9/L	5.91 (5, 7.1)	6.75 (5.51, 7.95)	6.28 (5.18, 7.84)
Red Blood Cell Count, 10^12/L	4.61 (4.31, 4.96)	4.64 (4.26, 5.03)	4.58 (4.26, 4.96)
Platelet Count, 10^9/L	224.5 (188, 259.25)	214 (186, 250.5)	225 (179, 253)
Hemoglobin, g/L	139 (130, 149)	141 (129.5, 154)	139 (126, 151)
Basophil Percentage, n (%)	0.4 (0.3, 0.6)	0.5 (0.33, 0.6)	N.A
Eosinophil Percentage, n (%)	2 (1.3, 3.1)	1.75 (1, 3.28)	N.A
MPV, fL	10.1 (9.4, 10.8)	9.25 (8.7, 10.2)	N.A
NLR	1.73 (1.32, 2.25)	2.26 (1.59, 2.73)	2.8 (2.1, 3.82)
Glucose, mmol/L	5 (4.69, 5.51)	5.53 (4.97, 6.11)	5.07 (4.47, 5.84)
Total Cholesterol, mmol/L	4.74 (4.1, 5.36)	4.74 (4.05, 5.36)	4.37 (3.5, 5.02)
HDL-C, mmol/L	1.32 (1.13, 1.51)	1.2 (0.99, 1.54)	1.03 (0.86, 1.27)
LDL-C, mmol/L	2.65 (2.15, 3.12)	2.71 (2.28, 3.35)	2.4 (1.83, 2.88)
ALT, U/L	18 (13, 25)	18 (13.5, 25)	19 (14, 30)
AST, U/L	19 (17, 23)	20 (17, 23.5)	18 (16, 25)
Uric Acid, μmol/L	315 (263.75, 371)	348.15 (300.13, 440.75)	386 (334, 437)
Triglycerides, mmol/L	1.16 (0.82, 1.70)	1.48 (1.07, 2.1)	1.56 (1.03, 2.04)

Center A: Jiangsu Province Hospital of Chinese Medicine; **Center B**: Suzhou Hospital of Integrated Traditional Chinese and Western Medicine; **Center C**: Zhongda Hospital, Southeast University School of Medicine.

Table 7: **Descriptive statistics of basic information and continuous feature of external validation dataset.**

Fig. 10. The prediction results are given as high risk or low risk for CKD, along with the corresponding percentage. Additionally, using the SHAP interpretability method, the importance of each feature is provided, with red indicating an increase in CKD risk and blue indicating a decrease in CKD risk, as shown in Fig. 11. For more information about the platform, see Supplementary Appendix S2.

Discussion

CKD is a progressive condition, with early stages often lacking noticeable symptoms, leading to a large number of high-risk individuals being undiagnosed in the early phases.[27] This study leverages large-scale, multi-center retrospective data to apply various ML models for early CKD prediction and to identify the most suitable combinations of diagnostic tests. The goal is to provide more accurate prediction tools for clinical practice, which not only improves early diagnosis rates of CKD but also offers new insights for personalized treatment and risk assessment.

ML has demonstrated significant advantages in disease prediction.[28] In this study, six different ML models were used to predict early-stage CKD. The experimental results show that gradient boosting tree models such as

Feature	0: Negative 1: +/± 2: ++,+++	Center A (n = 704)
Vitamin C	0	631/704 (89.63)
	1	18/704 (2.56)
	2	55/704 (7.81)
Ketones	0	685/704 (97.3)
	1	17/704 (2.41)
	2	2/704 (0.28)
Protein	0	666/704 (94.6)
	1	36/704 (5.11)
	2	2/704 (0.28)
Leukocytes	0	516/704 (73.3)
	1	132/704 (18.75)
	2	56/704 (7.95)
Occult Blood	0	419/704 (59.52)
	1	274/704 (38.92)
	2	11/704 (1.56)
Glucose	0	663/704 (94.18)
	1	8/704 (1.14)
	2	33/704 (4.69)
Bilirubin	0	701/704 (99.57)
	1	3/704 (0.43)
	2	N.A
Nitrite	0	697/704 (99.01)
	1	1/704 (0.14)
	2	6/704 (0.85)

Table 8: **Descriptive statistics of urinalysis categorical feature of external validation dataset.**

XGBoost and LightGBM exhibited excellent performance, particularly in handling multi-feature, high-complexity data. Their generalization ability and predictive performance far surpass traditional models.[29]

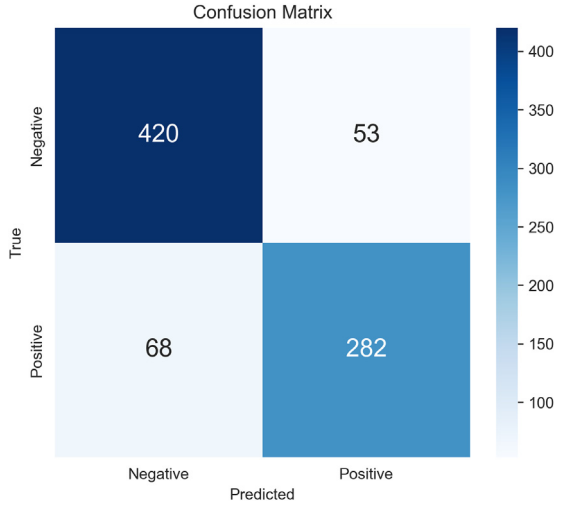


***Fig. 9:* Confusion Matrix of External Validation**. The matrix shows the distribution of true positives, true negatives, false positives, and false negatives. The model correctly identified 282 positive cases (CKD) and 420 negative cases (non-CKD), with 68 false negatives and 53 false positives.

XGBoost significantly outperformed other models. This suggests that the model not only captures the contribution of individual biomarkers to CKD risk but also identifies interactions between different features, allowing for more accurate early-stage CKD risk predictions. XGBoost, as an efficient gradient boosting algorithm, excels at handling nonlinear and high-dimensional sparse data. Its built-in regularization mechanism controls model complexity, reducing overfitting, while parallel computing accelerates the training process, ensuring high computational efficiency even with large datasets.[8] Additionally, the automation and scalability of these models make them highly advantageous in clinical applications. The flexibility of ML models allows them to handle incomplete or inconsistent data effectively, reducing prediction biases caused by missing or heterogeneous data.[30]

In multidimensional CKD prediction, feature selection is a key step in determining model performance. Identifying the most explanatory variables for disease risk from a large set of features has been a challenge in both clinical practice and data science.[31] This study used SHAP to quantitatively analyze each feature's contribution to the model's predictions.[32] SHAP values provide both global and local impact analysis for each feature, and the results show that proteinuria (Pro) plays a dominant role in predicting early-stage CKD risk, with its average SHAP value significantly higher than other features. This finding aligns with known renal physiology mechanisms, where creatinine is a critical indicator of kidney function decline and plays a decisive role in CKD prediction. Additionally, age (AGE) is also highlighted by SHAP values, suggesting its significant relevance in CKD progression prediction. This data-driven feature selection avoids biases from human assumptions and provides clinicians with an intuitive and reliable ranking of feature importance, which aids in optimizing diagnostic and intervention strategies. Moreover, SHAP values also reveal potential correlations between features, such as the non-linear interaction effects of the combination of HGB, BASO%, and MPV. Through SHAP value clustering analysis, we gain deeper insights into the complex relationships among these features, offering important clues for subsequent biological mechanism studies.

The XGBoost model demonstrated excellent performance in feature selection, model optimization, and handling high–dimensional complex data. By combining features from blood tests, urine tests, biochemical indicators, and basic information, the model achieved an AUC of 0.9143 and accuracy of 0.8561 (internal validation), as well as an AUC of 0.8962 and accuracy of 0.853 (external validation), showing high prediction accuracy and good generalization ability. The model not only effectively handles interactions between multiple features but also accurately predicts disease risk in the complex pathological context of CKD,

Youden' J	Threshold	Accuracy	F1 Score	Sensitivity	Specificity	PPV	NPV	AUC	PR-AUC
0.6937	0.4122	0.853	0.8234	0.8057	0.8879	0.8418	0.8607	0.8962	0.8878

ROC-AUC: Area Under the Receiver Operating Characteristic Curve; PR-AUC: Area Under the Precision-Recall Curve; PPV: Positive Predictive Value; NPV: Negative Predictive Value.

Table 9: **Model evaluation metrics for external validation.**

highlighting its robust ability to handle complex, multidimensional data.

This study also evaluated the impact of different feature combinations on model performance. The results indicate that the combination of "urine test + biochemical indicators + basic information" performed best, emphasizing the importance of urine and biochemical indicators in CKD risk assessment. This combination leverages key indicators of kidney function damage and can accurately distinguish patients with CKD from healthy individuals. However, biochemical tests are costly and may not be routinely available in some resource-limited medical settings. To address this, the study also evaluated other feature combinations. Notably, the "blood test + urine test + basic information" combination also demonstrated good predictive ability, although slightly lower accuracy compared to the combination including biochemical indicators. Nevertheless, it still provided high prediction accuracy and diagnostic value, particularly in situations where biochemical data

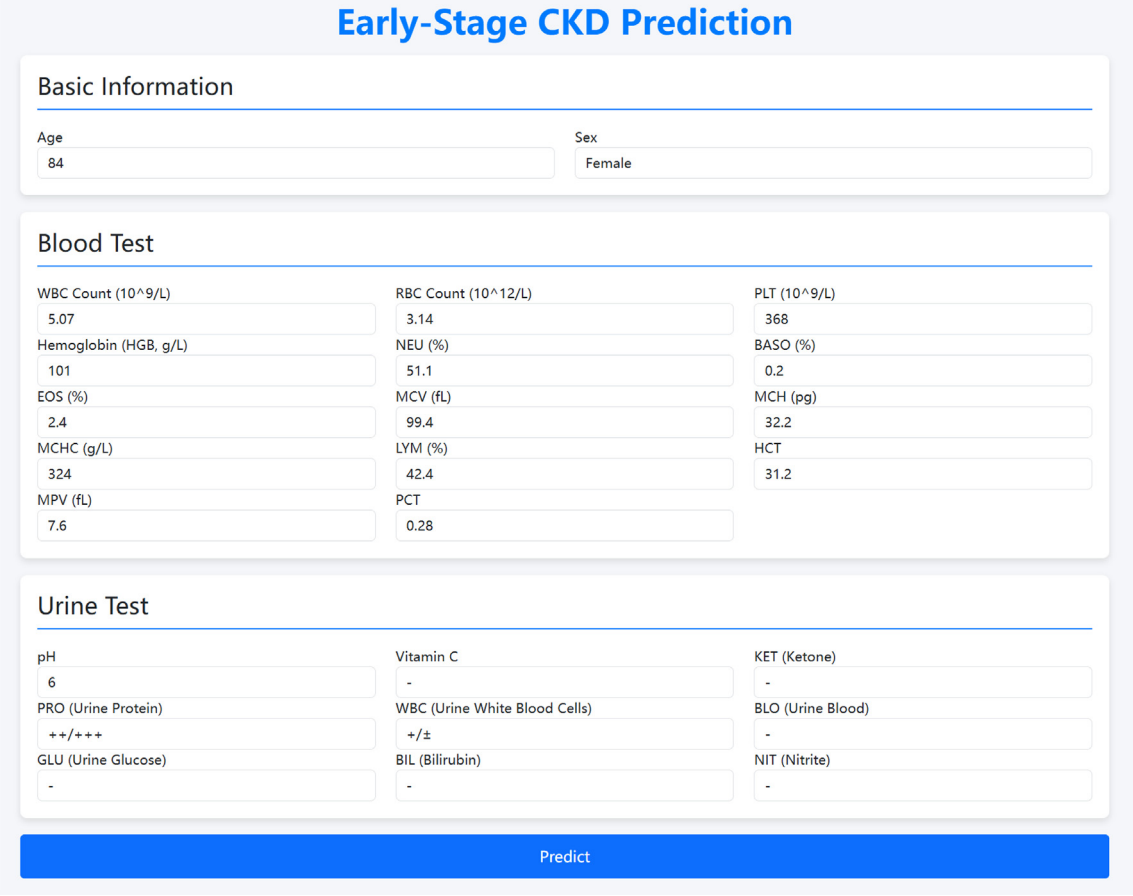


Fig. 10: **Web interface**. The interface is divided into three main sections: Basic Information, Blood Tests, and Urine Tests. The Basic Information section includes input fields for age and gender. The Blood Tests section includes input fields for common blood parameters such as White Blood Cell Count (WBC), Hemoglobin (HGB), Red Blood Cell Count (RBC), and Platelet Count (PLT). The Urine Tests section includes indicators like pH, Protein in Urine (PRO), and Glucose in Urine (GLU). After filling in all the required information, the user can click the "Predict" button to perform CKD risk prediction.

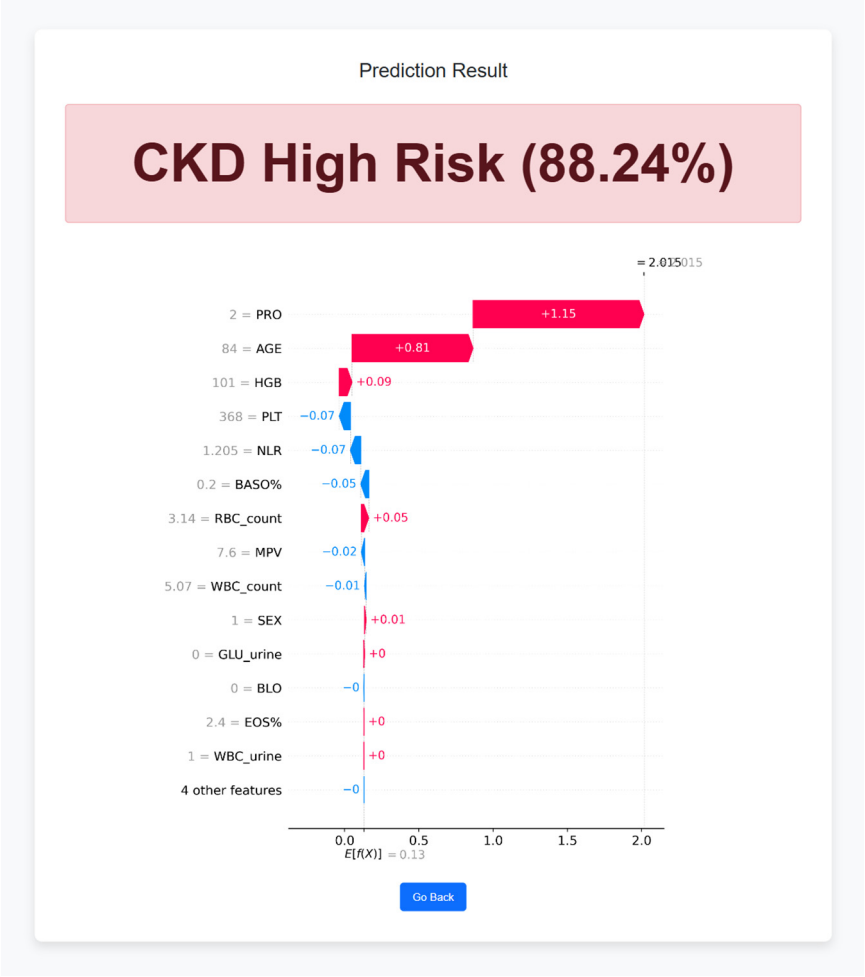


Fig. 11: **Prediction result**. The model predicts that the patient is at high risk for CKD, with a probability of 88.24%. The figure shows the contribution of each feature to the final predicted score. Red SHAP values are positive, indicating that the feature increases the CKD risk prediction value, while blue SHAP values are negative, indicating that the feature decreases the risk prediction value.

is unavailable. In these cases, the combination of blood and urine tests can serve as an effective alternative for early CKD prediction.

Further analysis revealed that the performance differences of the XGBoost model across various feature combinations were minimal, demonstrating strong robustness and generalization ability. This robustness indicates that the XGBoost model can maintain high prediction accuracy even with large-scale datasets or incomplete feature sets. This is especially important for CKD, a complex and slowly progressing disease, as it allows the model to adapt to different patient groups and provide consistent, reliable predictions.

The XGBoost model in this study also demonstrated excellent generalization ability in external validation. The model performed well in external validation datasets from three independent hospitals, indicating that it not only performed excellently on the local dataset but also has good applicability across different regions and hospitals. Notably, the heterogeneity of the external validation datasets enhanced the reliability of the model. In multi-center studies, data from different regions and hospitals often exhibit variations in sample characteristics, such as demographic features, examination methods, and patient disease progression, all of which can influence model prediction capability. Therefore, the model's strong performance on heterogeneous datasets further validates the robustness and practical applicability of the XGBoost model. In addition, the subgroup analysis results further support the applicability of the model across populations with different characteristics. In the vast majority of subgroups, the model maintained a high level of discriminative performance (Supplementary Table S3).

Despite the significant findings of this study, there are still some notable limitations that should be addressed. As a retrospective study, the sample sizes across the three cohorts exhibit a bias, with CKD data primarily derived from Center A and physical examination data mainly from Center B. Additionally, the physical examination data lacks descriptions of related conditions such as diabetes and hypertension. The time span of the data may also introduce heterogeneity between samples, potentially affecting the stability and generalizability of the model. Future studies should incorporate prospective data to further validate the model's predictive ability and expand its applicability to different populations and disease states. Additionally, with the further development of artificial intelligence technologies, powerful reasoning and decision-making capabilities, as well as knowledge emergence from large language models[33–35] and graph neural networks,[36–38] are expected to provide more accurate solutions for CKD prediction.

This study systematically analyzed multidimensional data of patients with CKD and validated the potential of ML models such as XGBoost and LightGBM in early-stage CKD prediction. Notably, the XGBoost model demonstrated outstanding predictive performance and broad applicability. Combined with SHAP-based interpretability analysis, the model's clinical application becomes more transparent and reliable, providing strong support for early-stage CKD detection.

To ensure dataset balance and minimize the impact of data partitioning on model training, the study employed multiple imputation, stratified random sampling, and K-fold cross-validation for data preprocessing. During model evaluation, in addition to conventional evaluation metrics, the DeLong method was introduced to compare the AUC of different models and assess statistical significance. Additionally, Youden's J index was calculated to balance model sensitivity and specificity. These methods were implemented to enhance model stability and robustness, ensuring the reliability of the study's conclusions. Although cross-validation was used to mitigate the risk of overfitting, the model may still exhibit some degree of overfitting due to the sample size and feature dimensionality limitations of the dataset. Future research should aim to reduce overfitting risk by expanding the sample size, particularly through further multicenter data collection, and by employing more rigorous model evaluation approaches, such as temporal validation. Furthermore, to address missing data on related conditions such as diabetes and hypertension, future studies will collect more comprehensive case data to ensure the completeness and reliability of model training. Future research should continue to explore feature selection and optimization strategies, integrate multimodal data[39] (such as imaging data,[40–42] proteomics,[43–45] metabolomics,[46,47] single-cell technologies,[48,49] etc.) to further improve the model's predictive capability and applicability. At the same time, with advancements in health monitoring technologies, real-time dynamic monitoring combined with ML models will open new directions for early warning and personalized treatment of CKD.[50]

Contributors

Jiayi He, Xin Wang performed the data analyses, established the ML models, and drafted the manuscript. *Peiqi Zhu, Xiaoxu Wang* participated in data collection, data analyses and confirmed the validity of the experiments. *Jing Zhao, Wei Sun, Kongfa Hu, Weiming He* participated in the design of the study and coordination. *Jiadong Xie* had primary responsibility for study design, data analyses, data interpretation, and writing the manuscript. *Jiadong Xie, Yitong Zhang, Peiqi Zhu* have accessed and verified the data. All authors read and approved the final manuscript.

Data sharing statement

The data analyzed and the codes used during the current study are available from the corresponding author on reasonable request.

Declaration of interests

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.eclinm.2025.103286.

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